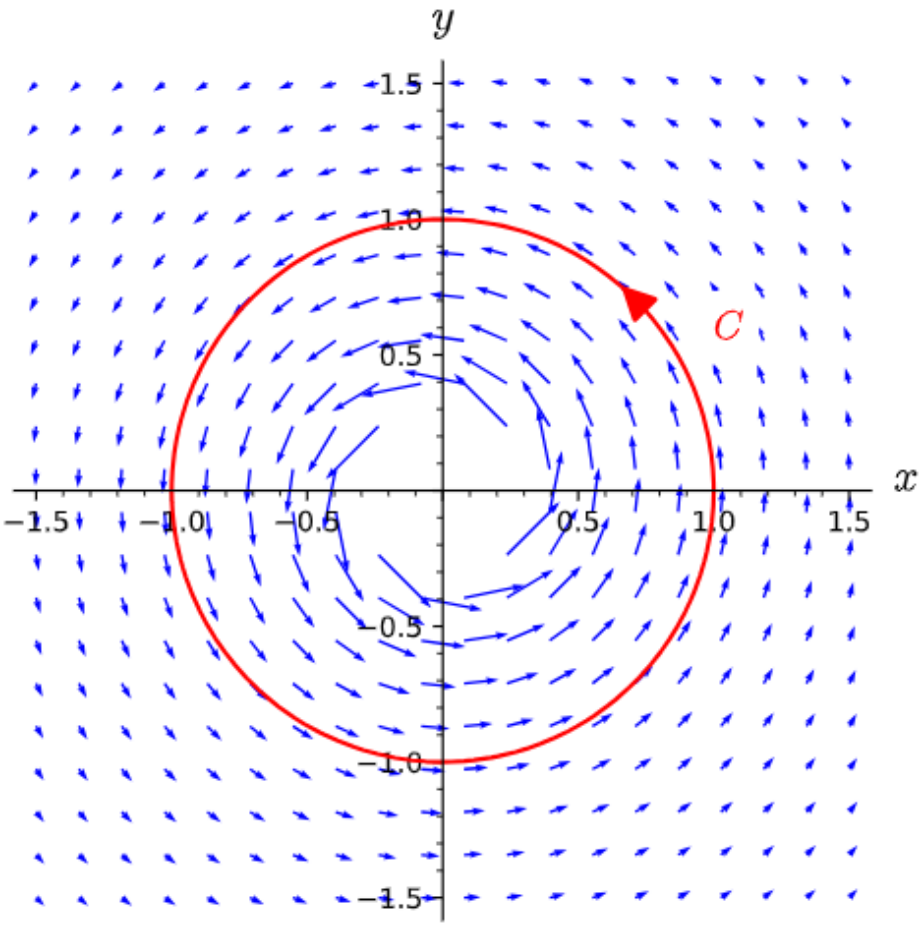
Activity 12.8.3. Consider the vector field $\mathbf{F} = \dfrac{-y}{x^2+y^2}\mathbf{i} + \dfrac{x}{x^2+y^2}\mathbf{j}$.

Notice that $\mathbf{F}$ is smooth everywhere in the plane other than at the point $(0,0)$. This vector field is plotted in Figure 12.8.6, but we have not plotted the vectors close to the origin as their magnitudes get so large that they make it hard to interpret the figure. Here Green's Theorem applies to any simple closed curve C that neither passes through $(0,0)$ nor bounds a region containing $(0,0)$.

Theorem 12.8.3. Green's Theorem. *Let C be a simple closed curve in the plane that bounds a region R with C oriented in such a way that when walking along C in the direction of its orientation, the region R is on our left. Suppose that $\mathbf{F} = \langle F_1, F_2 \rangle$ is vector field with continuous partial derivatives on the region R and its boundary C. The circulation of $\mathbf{F}$ along C may be calculated as*

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA.$$

in-context



From the above picture, it is clear that $\int_C \boldsymbol{F}(\boldsymbol{r})\cdot d\boldsymbol{r} > 0$ because the tangent vector along C is point to the same direction of vector field $\boldsymbol{F}$.

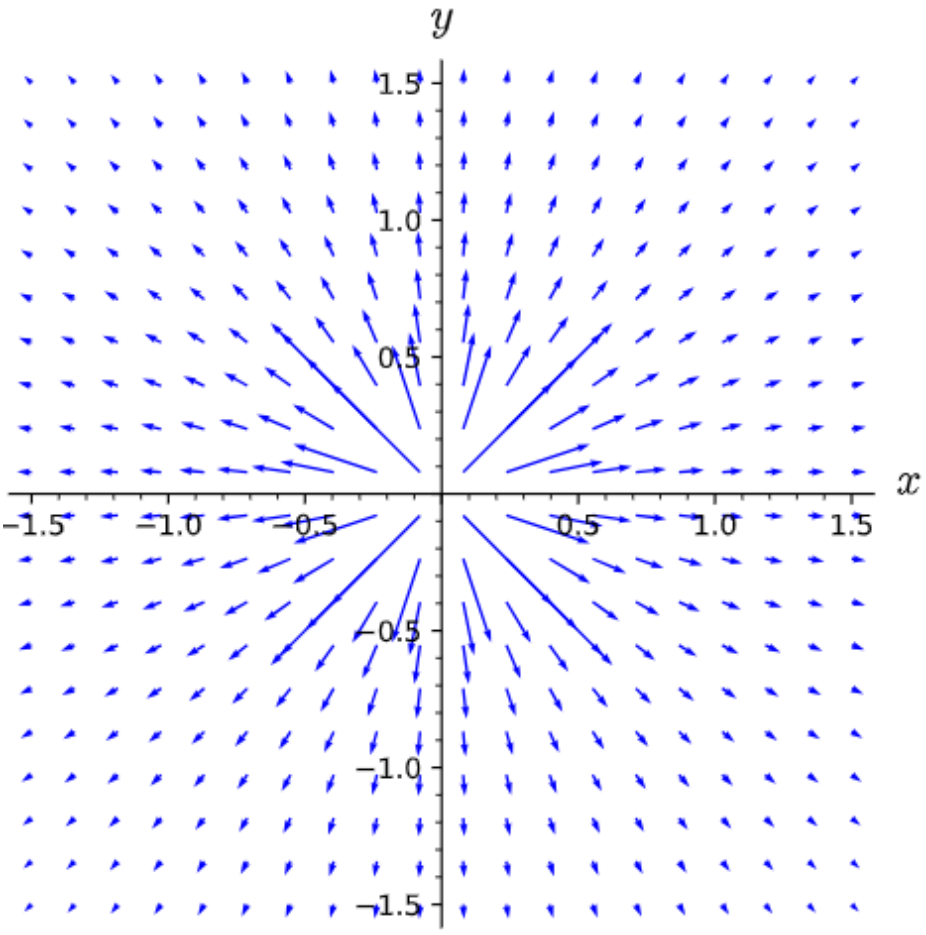
Let R be a bounded region on $\mathbb{R}^2$ with boundary C_1 (inner) and C_2 (outer) where C_1 and C_2 are closed curves traveling around 0 in positive direction.

$$\frac{\partial F_2}{\partial x}-\frac{\partial F_1}{\partial y}=\frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2}\right)-\frac{\partial}{\partial y}\left(\frac{-y}{x^2+y^2}\right)=\frac{(x^2+y^2)-2x^2}{(x^2+y^2)^2}+\frac{(x^2+y^2)-2y^2}{(x^2+y^2)^2}=0$$

$$0=\iint_{\partial R}\left(\frac{\partial F_2}{\partial x}-\frac{\partial F_1}{\partial x}\right)dxdy=\int_{C_2}\boldsymbol{F}(\boldsymbol{r})\cdot d\boldsymbol{r}-\int_{C_1}\boldsymbol{F}(\boldsymbol{r})\cdot d\boldsymbol{r}$$

$$\int_{C_2}\boldsymbol{F}(\boldsymbol{r})\cdot d\boldsymbol{r}=\int_{C_1}\boldsymbol{F}(\boldsymbol{r})\cdot d\boldsymbol{r}$$

Let $\boldsymbol{G}(x,y)=\left\langle\frac{x}{x^2+y^2},\frac{y}{x^2+y^2}\right\rangle$. Its graph of vector field is shown as below picture.



It is clear that $\int_C \boldsymbol{G}(\boldsymbol{r})\cdot d\boldsymbol{r}=0$ because the tangent vector along C is perpendicular to direction of vector field $\boldsymbol{G}$.

Domain D is not simply connected. It does not mean that the line integral of a conservative vector field along a closed curve C must be non-zero. $\int_C \boldsymbol{F}(\boldsymbol{r})\cdot d\boldsymbol{r}>0$, $\int_C \boldsymbol{G}(\boldsymbol{r})\cdot d\boldsymbol{r}=0$. It depends on vector field and C.